

KSA Office IT Support & Infrastructure Management

Phase 07: IT Support Operations

Afaq Business Solutions Co.

Log Analysis & Incident Investigation Standard Operating Procedure (SOP)

1. Purpose

Provide a structured method for investigating application incidents through log analysis, identifying likely causes, applying controlled remediation, and documenting the outcome.

2. Scope

For IT Support personnel investigating application failures, recurring errors, unavailable dependencies, and similar incidents using sanitized application logs.

3. Incident Investigation Workflow

- Document the user's reported symptoms and assign a ticket.
- Review the relevant application log.
- Filter ERROR entries.
- Review surrounding log entries to establish event sequence.
- Count recurring errors when useful.
- Identify the likely failed component or dependency.
- Apply controlled and authorized remediation.
- Verify the corrective configuration or dependency.
- Document findings, resolution, and user communication.
- Close the ticket after verification.

4. Log Review Commands

- Use **cat** to review a sanitized application log.
- Use **grep** to isolate ERROR entries.
- Use **grep -n -C 1** to inspect context around errors.
- Use **grep -c** to count recurring error patterns.

5. Root-Cause Analysis

- Establish the sequence from application startup through the failure.
- Distinguish application startup problems from dependency or data-source problems.
- Use repeated error patterns as supporting evidence rather than assuming the first error is the root cause.

6. Controlled Remediation

- Use sanitized lab configuration for training exercises.
- Restore only requires configuration values.
- Avoid modifying production systems without authorization.
- Never expose credentials, tokens, connection strings, or secrets.

7. Verification and Security

- Verify restored configuration with an appropriate read-only command.
- Store evidence in the designated Evidence directory.
- Sanitize usernames, hostnames, IP addresses, credentials, tokens, and other sensitive data before sharing screenshots.
- Document verification before closing the ticket.

8. Escalation

- Escalate incidents involving infrastructure, database availability, application defects, or changes outside support authorization.
- Include the relevant error sequence and supporting evidence when escalating.

9. Best Practices

- Use evidence before making changes.
- Prefer read-only diagnostics during initial investigation.
- Make the smallest authorized corrective change.
- Verify after remediation.
- Maintain clear incident documentation.